Prototype 1 – Reverse-Projection Validation

Blender + OpenCV PnP Forensic Workflow

Robert Bryan · October 2025

Overview

This document demonstrates a validated reverse-projection pipeline integrating Blender's camera-solving environment with OpenCV's Perspective-n-Point (PnP) solver. The goal was to verify that the workflow can recover accurate camera intrinsics and extrinsics — including focal length, optical center, and orientation — using imagery alone, and to quantify accuracy via reprojection error metrics.

Reverse-projection is a forensic photogrammetric technique used to determine real-world measurements, camera positions, or subject locations from images or video. Commercial forensic software such as iNPUT-ACE and ARAS employ similar PnP-based camera models, though their calibration workflows are proprietary. This work presents one of the first openly documented, reproducible implementations of these principles using publicly available tools built on open-source computer-vision frameworks such as OpenCV.

The approach leverages a scaled 3D photogrammetry model of Alcatraz Island (175k-triangle mesh sourced from Sketchfab) and a set of 2D–3D reference correspondences to reconstruct the camera's pose and lens characteristics. Through two controlled validation tests — a closed-loop render and an open-loop image — the pipeline achieved near sub-pixel reprojection accuracy under known conditions and consistent alignment when solving unknown imagery.

Discussion

These two tests together confirm that the Blender + OpenCV workflow can accurately recover camera pose and focal length from image data. The closed-loop result (1.04 px RMS) verifies mathematical soundness, corresponding to an approximate spatial margin of error of ±1.4 cm at a 120 m distance. The open-loop test (5.16 px RMS) demonstrates robustness under less-constrained, real-world conditions, with a corresponding spatial margin of error of roughly ±7 cm. Together, these results validate that the reverse-projection pipeline performs reliably across both controlled and unknown scenarios, maintaining measurable accuracy within acceptable forensic error thresholds. Margin-of-error values were calculated using standard pinhole projection geometry, verified through OpenCV calibration documentation, and cross-checked for numerical consistency to the best knowledge of the author.

Closed-Loop Validation

This test validated the pipeline's internal accuracy using a synthetic reference image rendered from a known camera pose. The same image was reintroduced into the PnP solver to test whether the workflow could recover the original camera parameters without prior knowledge. The solver successfully reconstructed the focal length, optical center, and orientation with an RMS reprojection error of 1.04 pixels across 32 reference points, confirming precise geometric consistency.

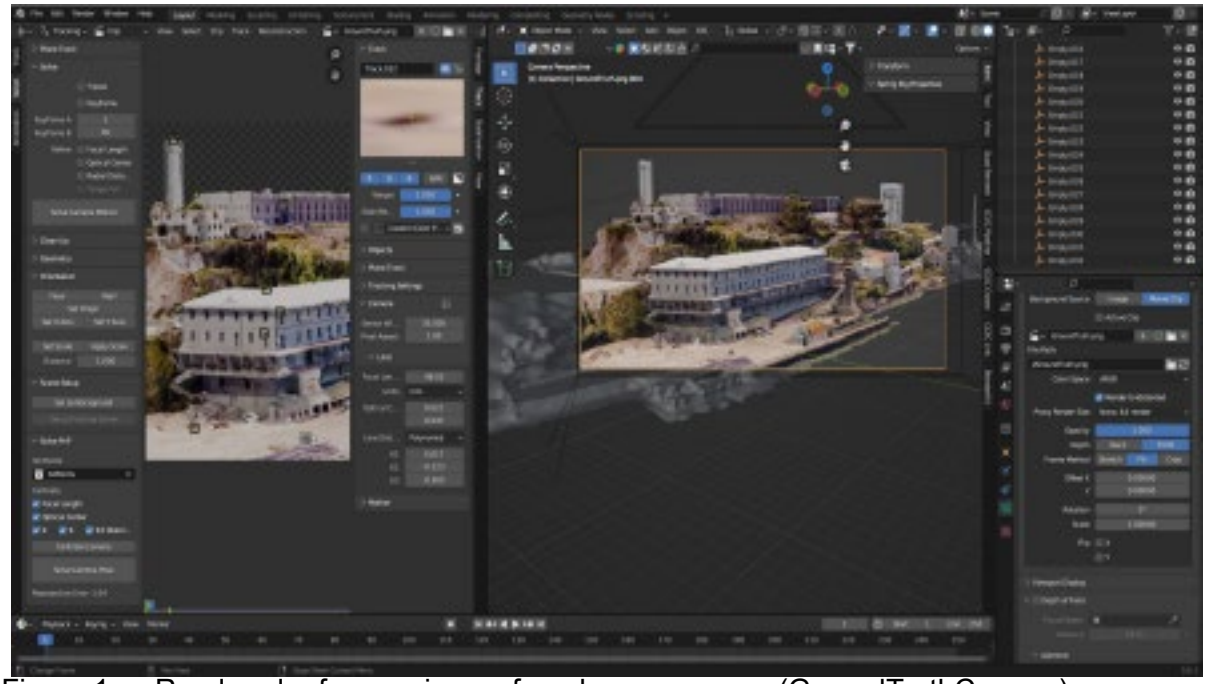

Figure 1 — Rendered reference image from known camera (GroundTruthCamera).

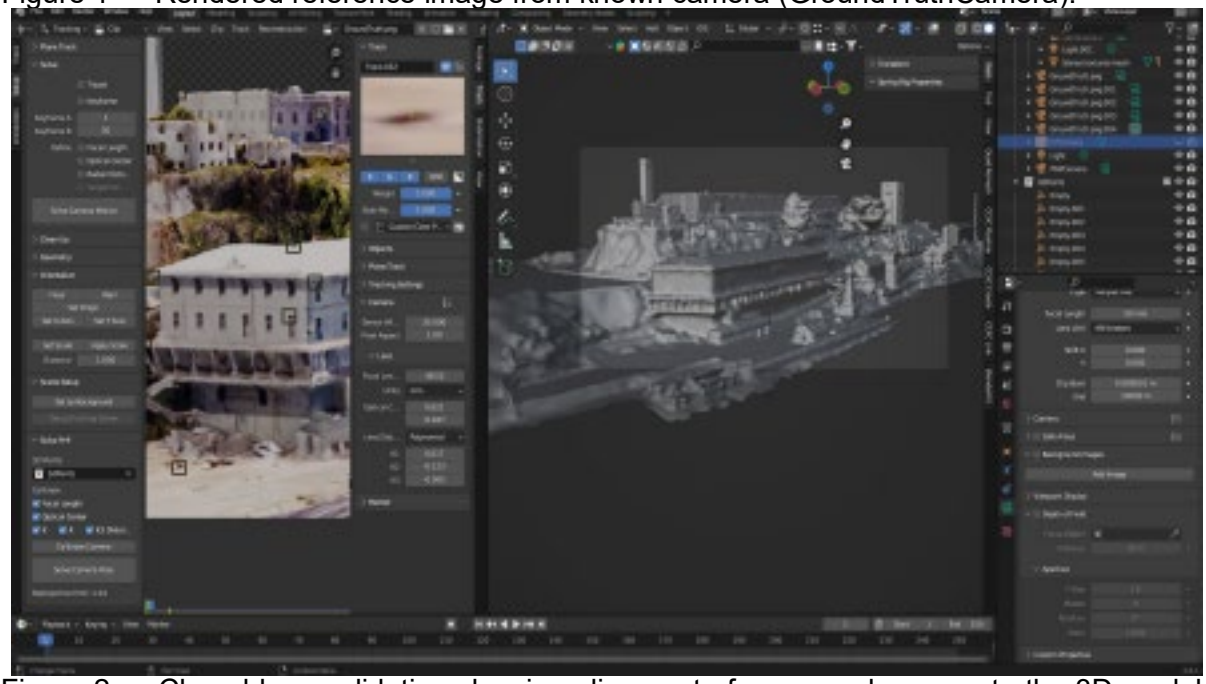

Figure 2 — Closed-loop validation showing alignment of recovered camera to the 3D model (RMS 1.04 px).

Open-Loop Validation

This second test evaluated the workflow using an external image of Alcatraz Island with unknown camera intrinsics, simulating real-world conditions. The solver was allowed to estimate focal length, optical center, and radial distortion while aligning the 3D model to the photograph. Despite the low-resolution photogrammetry data, the solver achieved an RMS reprojection error of 5.16 pixels across 32 reference points, demonstrating strong alignment stability even without prior calibration data.

Figure 3 — Open-loop validation using an external reference image

Figure 4 — Solved camera alignment showing consistent registration to the model (RMS 5.16 px).

Conclusion

To the best knowledge of the author, this work presents one of the first openly documented and reproducible implementations of these principles using publicly available tools built on open-source computer-vision frameworks such as OpenCV. While commercial forensic

software such as iNPUT-ACE and FARO/ARAS 360 rely on similar PnP-based camera models, their underlying calibration methods remain proprietary. The workflow presented here provides a transparent and auditable alternative for validation, instruction, and research. Prototype 1 demonstrates a fully functional, quantitatively verified reverse-projection system that integrates accurate scaling, camera calibration, and reprojection analysis—establishing a solid foundation for future forensic-grade reconstructions using higher-fidelity LiDAR or multi-view datasets.

References & Resources

Open-Source Tools Used

Blender — https://www.blender.org

OpenCV (Open Source Computer Vision Library) — https://opencv.org

OpenCV: Perspective-n-Point (PnP) Solver Documentation —

https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html

camera-pnpoint https://github.com/RT-studios/camera-pnpoint/

3D Data Source

Alcatraz Island Photogrammetry Model (Sketchfab) — https://sketchfab.com/3d-models/alcatraz-island-86a8cac8621e4403882cf1dd6ab0f321

Related Forensic Software (Referenced for Context)

iNPUT-ACE Video Evidence Analysis Suite — https://www.axon.com/products/axon-investigate

ARAS 360 / ARAS 360 HD — https://aras360.com